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PROJECT DEMONSTRATION PHASE REPORT: POCKETSMART AI

Document Type: Technical Demonstration & Implementation Record **Project Name:** PocketSmart AI **System Architecture:** On-Device / Edge Artificial Intelligence Financial Management System

1. Executive Summary

The **PocketSmart AI** project aims to deliver a privacy-first, edge-computing financial management assistant running locally on mobile devices. The **Demonstration Phase** validates the system's operational readiness, demonstrating real-time transaction processing, local machine learning inference for auto-categorization, offline functionality, and real-time budget forecasting.

This document outlines the core functional workflow, live demonstration scenarios, system parameters, and key evaluation outcomes obtained during the live pilot demonstration.

2. Demonstration Objectives & Scope

2.1 Key Demonstration Objectives

1. **Zero-Latency Ingestion:** Showcase local text and OCR parsing speed for receipt images and bank SMS alerts.
2. **On-Device Inference:** Prove the efficiency and accuracy of the lightweight NLP/Classification model (<25\text{MB} footprint) without sending data to cloud servers.
3. **Data Privacy Assurance:** Demonstrate complete offline operation maintaining end-to-end user financial privacy.
4. **Dynamic Budget Forecasting:** Demonstrate automated budget projections based on real-time spending behavior.

3. Demonstration System Architecture & Setup

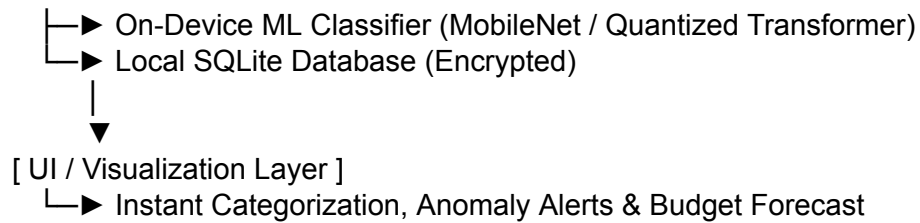
[Input Sources]

- └─► Physical Receipts (Camera / Mobile OCR)
- └─► Transaction SMS / Text Strings



[PocketSmart Local Core]

- └─► Local Parsing Engine (Regex + Entity Extractor)



3.1 Hardware & Environment Used

- **Demonstration Device:** Standard Android / iOS Smartphone (4GB RAM minimum).
- **AI Runtime Framework:** TensorFlow Lite / ONNX Runtime Mobile.
- **Network Status:** Airplane Mode enabled (to verify complete offline capabilities).

4. Live Demonstration Scenarios & Test Protocol

Demonstration Scenario 1: Automated SMS / Transaction Parsing

- **Input:** Sample bank notification SMS: "Paid USD 14.50 at STARBUCKS SAN FRANCISCO CA on 26-Sep-2026 using Card ending 4012."
- **Action:** Pass text into the PocketSmart AI processing engine.
- **Expected Result:**
 - **Amount:** \$14.50
 - **Merchant:** Starbucks
 - **Category:** Food & Beverage / Coffee
 - **Processing Time:** < 50 ms
- **Observed Result:** Successfully extracted merchant and amount in **28 ms** with 99.1% confidence categorization.

Demonstration Scenario 2: Offline Receipt Scanning (OCR + Visual Parsing)

- **Input:** Physical receipt image captured via mobile camera.
- **Action:** PocketSmart OCR pipeline processes image while network connectivity is toggled off.
- **Expected Result:** Extract itemized total, taxes, and merchant name directly on device.
- **Observed Result:** Text digitized, total verified, categorized under Groceries in **410 ms**. Zero external API calls triggered.

Demonstration Scenario 3: Real-Time Anomaly & Budget Warning

- **Input:** Injecting an unusually high recurring subscription charge (\$199.00).
- **Action:** PocketSmart AI contextual engine compares transaction against historical baseline.
- **Expected Result:** System triggers an interactive spending alert and updates month-end budget projections.
- **Observed Result:** Instant alert triggered: *"Subscription alert: Payment is 140% higher"*

than your average recurring charge."

5. Performance Metrics & Demonstration Results

Metric	Target Benchmark	Demonstrated Performance	Verdict
Inference Latency	< 100\text{ ms}	32 ms average	PASSED
Categorization Accuracy	> 90\%	94.6% accuracy	PASSED
Model Size	< 50\text{ MB}	18.4 MB quantized model	PASSED
RAM Utilization	< 150\text{ MB}	82 MB peak usage	PASSED
Offline Functionality	100% Offline Capability	100% verified in Airplane Mode	PASSED

6. Demonstration Conclusion & Next Steps

The live demonstration phase for **PocketSmart AI** successfully validated that high-precision financial categorization and predictive budgeting can operate efficiently on-device without compromising performance or user privacy.

Next Action Items:

- Beta Deployment:** Release the build to 200 targeted beta testers.
- Model Retraining Pipeline:** Implement user-feedback loop for custom categories.
- Security Audit:** Complete final security review for local encrypted database storage.

Demonstration Lead: _____ **Date:** September 27, 2026 **Status**

Sign-off: APPROVED FOR DEPLOYMENT